

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

○



[Lab.1] Surface Structures/ Surface Energies	 3
[Lab.2] Electronic Structure depend on number of layers	 26
[Lab.3] Oxygen Reduction Reaction	 49
[Lab.4] CO ₂ Reduction Reaction	 82
[Lab.5] Nitrogen Reduction Reaction	 105

[Lab.2] Electronic Structure depend on number of layers

Learning Objectives:

→ Create surface structures and check for changes in surface structure through structural optimization calculations. Also compare the difference in electronic structure as a function of the number of layers in the slab structure.

Materials: Pt(111) surfaces



MatSQ Credit: ~ 12 CPU hour (\$ 3.0)

[Lab.2] Electronic Structure depend on number of layers

Pt(111) surfaces

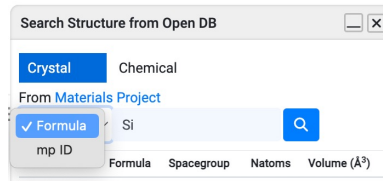
1. Build crystal structure
2. Calculate variable cell relaxation of bulk structure (primitive cell)
3. Build conventional cell with optimized lattice constants
4. Calculate total energy of Pt conventional cell
5. Build a Pt(111) surface structure with 6 layers and optimize the atomic structure
6. Build a Pt(111) surface structure with 4 layers and optimize the atomic structure
7. Calculate electronic structures
8. Analyze electronic structures

MatSQ Credit: ~ 12 CPU hour (\$ 3.0)

[Lab.2] Electronic Structure depend on number of layers

1. Build crystal structure (DB, Load File, Preset)

① Using OpenDB

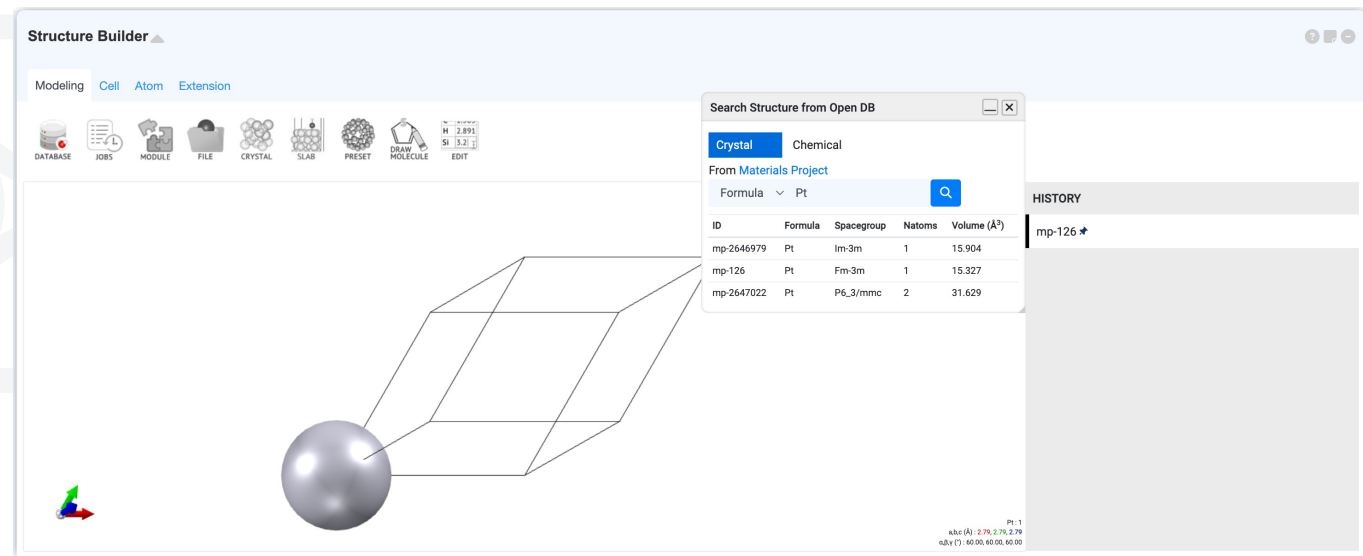
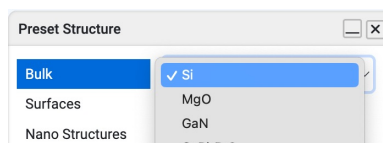


② Load files



.pdb .cif .xyz
...

③ Preset



[Lab.2] Electronic Structure depend on number of layers

2. Calculate variable cell relaxation of bulk structure (primitive cell)

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Bulk

Ionic+cell optimization

☐ Spin polarization
☐ Van der Waals
☐ Electric field
☒ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

vc-relax

Max SCF steps

200

Information amount

high

Force threshold (Ry/bohr)

0.00038

&IONS

Ion dynamics

bfgs

Upscale

100

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$

40

$E_{\text{cut}}(\text{rho})(\text{Ry})$

320

Occupations

smearing

Gaussian broadening (Ry)

0.002

&CELL

Cell dynamics

bfgs

Cell factor

2

Pressure threshold (Kbar)

0.2

Cell relax type

all

&ELECTRONS

Max iteration step

200

SCF must converge

True

Mixing mode

plain

Mixing beta

0.5

Convergence threshold (Ry)

0.000001

Starting wavefunction

random

KPOINTS

Sampling

Monkhorst-Pack

Grid

11 11 11

Shift

0 0 0

[Lab.2] Electronic Structure depend on number of layers

3. Build conventional cell with optimized lattice constant

Structure Builder

Modeling **Cell** Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Unit: Angstrom(Å)

Cell: Lattice Parameters

a	2.809534623	α (°)	60
b	2.809534623	β (°)	60
c	2.809534623	γ (°)	60

Atoms: Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
1	Pt	0	0	0

[+ Add Atom](#)

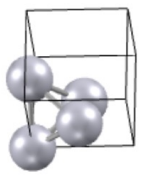
[Lab.2] Electronic Structure depend on number of layers

3. Build conventional cell with optimized lattice constant

Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL



HISTORY

- mp-126
- 147104-Final
- Manual Edit
- Conventional

Pt : 4
a,b,c (Å) : 3.97, 3.97, 3.97
 α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.2] Electronic Structure depend on number of layers

4. Calculate total energy of Pt conventional cell

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Bulk

No optimization

☐ Spin polarization

☐ Van der Waals

☐ Electric field

☐ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

Information amount

Force threshold (Ry/bohr)

scf

high

0.00038

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

40

320

smearing

0.002

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

200

True

plain

0.5

0.000001

random

Monkhorst-Pack

8 8 8

0 0 0

[Lab.2] Electronic Structure depend on number of layers

5-1. Build Pt(111) surface structure with 6 layers

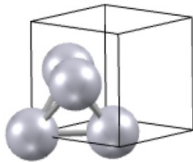
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch **THIS** to exit select mode

Select mode - Rectangular Normal



Cleave Surface

(h,k,l) = 1 1 1

(max c length Å) 30 **✓ Apply**

no.	a	b	c	α	β	γ
1	5.619	5.619	5.619	60.000	60.000	60.000
2	5.619	5.619	9.733	73.221	90.000	60.000
3	5.619	5.619	13.764	90.000	90.000	60.000
4	5.619	5.619	18.636	90.000	98.671	60.000
5	5.619	5.619	23.168	83.035	90.000	60.000
6	5.619	5.619	28.095	84.261	95.739	60.000

a,b,c (Å) : 3.97, 3.97, 3.97
 α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.2] Electronic Structure depend on number of layers

5-1. Build Pt(111) surface structure with 6 layers

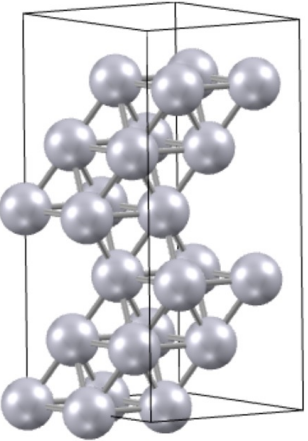
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal



History

- New Structure
- Cleave (1 1 1)

Pt : 24
a,b,c (Å) : 5.62, 5.62, 13.76
α,β,γ (°) : 90.00, 90.00, 60.00

[Lab.2] Electronic Structure depend on number of layers

5-1. Build Pt(111) surface structure with 6 layers

Structure Builder

Modeling **Cell** Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Unit: Angstrom(Å)

Cell: Lattice Parameters

a: 5.6190615219 α (°): 90

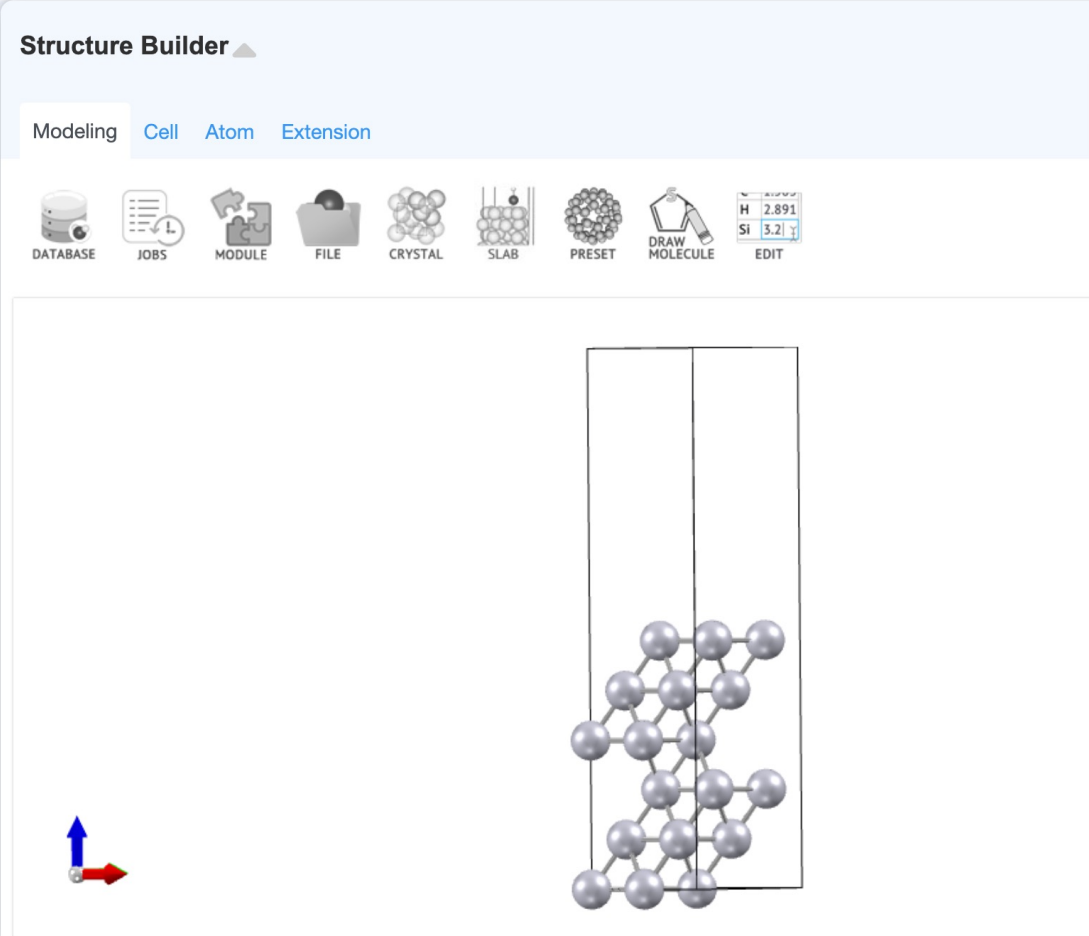
b: 5.6190615219 β (°): 90

c: 25 γ (°): 59.999999999

Atoms: Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
1	Pt	0	0	0
2	Pt	5.6190	3.2441	2.2939
3	Pt	2.8095	1.6220	4.5879
4	Pt	0	0	6.8819
5	Pt	5.6190	3.2441	9.1758

+ Add Atom



[Lab.2] Electronic Structure depend on number of layers

5-2. Optimize atomic structure of Pt(111) surface structure with 6 layers

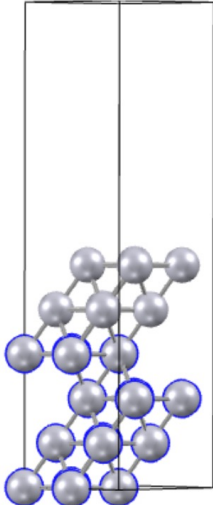
Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular
Normal
Pt=16



HISTORY

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt: 24
a,b,c (Å): 5.62, 5.62, 25.00
α,β,γ (°): 90.00, 90.00, 60.00

[Lab.2] Electronic Structure depend on number of layers

5-2. Optimize atomic structure of Pt(111) surface structure with 6 layers

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info

Target property
Energy / Structure

&CONTROL
Calculation type
scf

input.pw.x

Potential

&CONTROL
calculation = 'scf'
outdir = './output/'
prefix = 'VLAB'

Checkbox	Input	Selected
Fix		
No.	Ele.	☐ ☐ x ☐ y ☐ z
1	Pt	☒ 0.000 ☒ 0.000 ☒ 0.000
2	Pt	☒ 5.619 ☒ 3.244 ☒ 2.294
3	Pt	☒ 2.810 ☒ 1.622 ☒ 4.588
4	Pt	☒ 0.000 ☒ 0.000 ☒ 6.882
5	Pt	☐ 5.619 ☐ 3.244 ☐ 9.176
6	Pt	☐ 2.810 ☐ 1.622 ☐ 11.470
7	Pt	☒ 2.810 ☒ 0.000 ☒ 0.000
8	Pt	☒ 2.810 ☒ 3.244 ☒ 2.294
9	Pt	☒ 5.619 ☒ 1.622 ☒ 4.588

Structure optimization
No optimization

&ELECTRONS
Max iteration step
30
smearing
0.002
SCF must converge
True
Mixing beta
0.5
KPOINTS
Sampling
Monkhorst-Pack
Grid
3 3 1
Shift
0 0 0

Submit
Resource : Spot
PPN : 48 ram : 192 GB

[Lab.2] Electronic Structure depend on number of layers

5-2. Optimize atomic structure of Pt(111) surface structure with 6 layers

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure

Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic optimization

☐ Spin polarization

☐ Van der Waals

☐ Electric field

☐ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

relax

200

high

0.00038

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

40

320

smearing

0.002

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

200

True

plain

0.5

0.000001

random

Monkhorst-Pack

5

5

1

0

0

0

&IONS

Ion dynamics

Upscale

bfgs

100

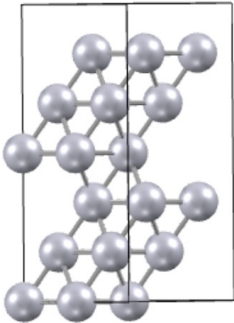
[Lab.2] Electronic Structure depend on number of layers

6-1. Build Pt(111) surface structure with 4 layers

Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT



Structure List

Name	Pt-bulk	
Formula	Pt4	
a,b,c(Å)	3.973 3.973 3.973	
α,β,γ (°)	90.000 90.000 90.000	
Name	Pt111-6layer	
Formula	Pt24	
a,b,c(Å)	5.619 5.619 25.000	
α,β,γ (°)	90.000 90.000 60.000	

HISTORY

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt : 24
G : 13.76
J0 : 60.00

[Lab.2] Electronic Structure depend on number of layers

6-1. Build Pt(111) surface structure with 4 layers

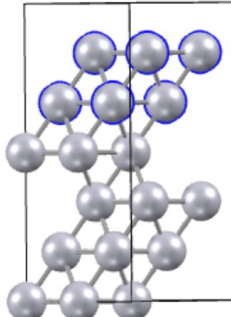
Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular
Normal
Pt=8



History:

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt: 24
a,b,c (Å): 5.62, 5.62, 13.76
α,β,γ (°): 90.00, 90.00, 60.00

[Lab.2] Electronic Structure depend on number of layers

6-1. Build Pt(111) surface structure with 4 layers

Structure Builder

Modeling **Cell** Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal

Unit: Angstrom(Å)

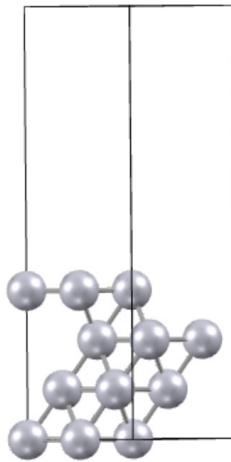
Cell: Lattice Parameters

a	5.6190615219	α (°)	90
b	5.6190615219	β (°)	90
c	20	γ (°)	59.999999999

Atoms: Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
1	Pt	0	0	0
2	Pt	5.6190	3.2441	2.2939
3	Pt	2.8095	1.6220	4.5879
4	Pt	0	0	6.8819
5	Pt	2.8095	0	0

+ Add Atom



[Lab.2] Electronic Structure depend on number of layers

6-2. Optimize atomic structure of Pt(111) surface structure with 4 layers

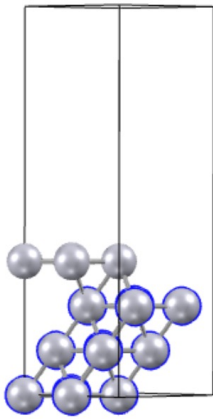
Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch **THIS** to exit select mode

Select mode - Rectangular
Normal
Pt=12



HISTORY

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt : 16
a,b,c (Å) : 5.62, 5.62, 20.00
α,β,γ (°) : 90.00, 90.00, 60.00

[Lab.2] Electronic Structure depend on number of layers

6-2. Optimize atomic structure of Pt(111) surface structure with 4 layers

Quantum Espresso

PWscf + DOS ▼

Scripting Option : Template ▼ Atom info ⚙

Target property
Energy / Structure

&CONTROL

Calculation type
scf

input.pw.x

Potential

&CONTROL
calculation = 'scf'
outdir = './output/'
prefix = 'VLAB'

	Checkbox	Input	Selected
Fix			
No.	Ele.	☐	☐ x ☐ y ☐ z
1	Pt	☒	☒ 0.000 ☒ 0.000 ☒ 0.000
2	Pt	☒	☒ 5.619 ☒ 3.244 ☒ 2.294
3	Pt	☒	☒ 2.810 ☒ 1.622 ☒ 4.588
4	Pt	☐	☐ 0.000 ☐ 0.000 ☐ 6.882
5	Pt	☒	☒ 2.810 ☒ 0.000 ☒ 0.000
6	Pt	☒	☒ 2.810 ☒ 3.244 ☒ 2.294
7	Pt	☒	☒ 5.619 ☒ 1.622 ☒ 4.588
8	Pt	☐	☐ 2.810 ☐ 0.000 ☐ 6.882
9	Pt	☒	☒ 1.405 ☒ 2.422 ☒ 0.000

Structure optimization
No optimization ▼

&ELECTRONS

Max iteration step
100

SCF must converge
True ▼

Mixing beta
0.5

KPOINTS

Sampling
Monkhorst-Pack ▼

Grid
3 3 1

Shift
0 0 0

Job Submit

Resource : Spot ▼

PPN : 48 ▼ ram : 192 GB

[Lab.2] Electronic Structure depend on number of layers

6-2. Optimize atomic structure of Pt(111) surface structure with 4 layers

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure

Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic optimization

☐ Spin polarization

☐ Van der Waals

☐ Electric field

☐ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

relax

200

high

0.00038

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

40

320

smearing

0.002

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

200

True

plain

0.5

0.000001

random

Monkhorst-Pack

551

000

&IONS

Ion dynamics

Upscale

bfgs

100

[Lab.2] Electronic Structure depend on number of layers

7. Calculate electronic structures

Quantum Espresso

PWscf (restart) DOS (1) + DOS ▼

Scripting Option : Template ▼ Atom info Update Structure Keyword information

Target property
DOS ▼

Precision
High (long time) ▼

Model type ?
Isolate System ▼

Structure optimization
No optimization ▼

Option
☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type	nscf ▼
Information amount	high ▼
Force threshold (Ry/bohr)	0.00038

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$	40
$E_{\text{cut}}(\text{rho})(\text{Ry})$	320
Occupations	smearing ▼
Gaussian broadening (Ry)	0.002

&ELECTRONS

Max iteration step	200			
SCF must converge	True ▼			
Mixing mode	plain ▼			
Mixing beta	0.5			
Convergence threshold (Ry)	0.000001			
Starting wavefunction	random ▼			
KPOINTS				
Sampling	Monkhorst-Pack ▼			
Grid	<table><tr><td>11</td><td>11</td><td>1</td></tr></table>	11	11	1
11	11	1		
Shift	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0
0	0	0		

[Lab.2] Electronic Structure depend on number of layers

7. Calculate electronic structures

Quantum Espresso

[PWscf \(restart\)](#) DOS (1) + DOS

&PROJWFC

ngauss	0
degauss	0.007349864
DeltaE	0.02
Emin	-20
Emax	40

[input.pw.x](#) [input.projwfc.x](#) [Potential](#)

```
&CONTROL
calculation = 'nscf'
forc_conv_thr = 0.00038
outdir = './output/'
prefix = 'VLAB'
pseudo_dir = './'
restart_mode = 'restart'
verbosity = 'high'
wf_collect = .TRUE.
/

&SYSTEM
degauss = 0.002
```

Job Submit

Resource : Spot

PPN : 48 ram : 192 GB

Node type : Computing

Job Name : 6layer-dos

Version : 6.4

Finish Notice : ☐ E-mail

Start Job!

[Lab.2] Electronic Structure depend on number of layers

8. Analyze electronic structures

